

Shri shamrao patil (yadravkar) Educational & Charitable Trust



Sharad Institute of Technology College Of Engineering

(An Autonomous Institute)

Department of Basic Sciences & Humanities

A

Mini Project Report

On

“Water Level

Indicator”

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Under the guidance of

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CERTIFICATE

This is certify that,

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Studying in F.Y. (Electronics and computer Engineering) has successfully completed the project entitled "Water Level Indicator" under the guidance and supervision of Miss. Purva Nikam during the academic year 2023-2024 in Basic Sciences and Humanities, SITCOE Yadav.

Miss. Purva Nikam

Dr. Sarika Patil

Dr. S.A.Khot

Mini Project Guide

HOD,BSH Department

Principal

ACKNOWLEDGEMENT

We take this opportunity to express our profound gratitude and deep regards to our guide Mrs. Swati Halunde & Miss. Purva Nikam for her exemplary guidance, monitoring and constant encouragement throughout the course the course of this project. The blessing, help and guidance given by her time to time shall carry us a long way in the journey of life on which we are about to embark.

We also take this opportunity to express a deep sense of gratitude to our principal, Dr. S. A. Khot and our HOD, Dr. Sarika Patil for their cordial support, valuable information and guidance, which helped us in completing this task through various stages.

We are obliged to all faculty and staff members of all departments of our Sharad Institute of Technology, College of Engineering, Yadrav for the valuable information provided by them in their respective fields. We are grateful for their cooperation during the period of our assignment.

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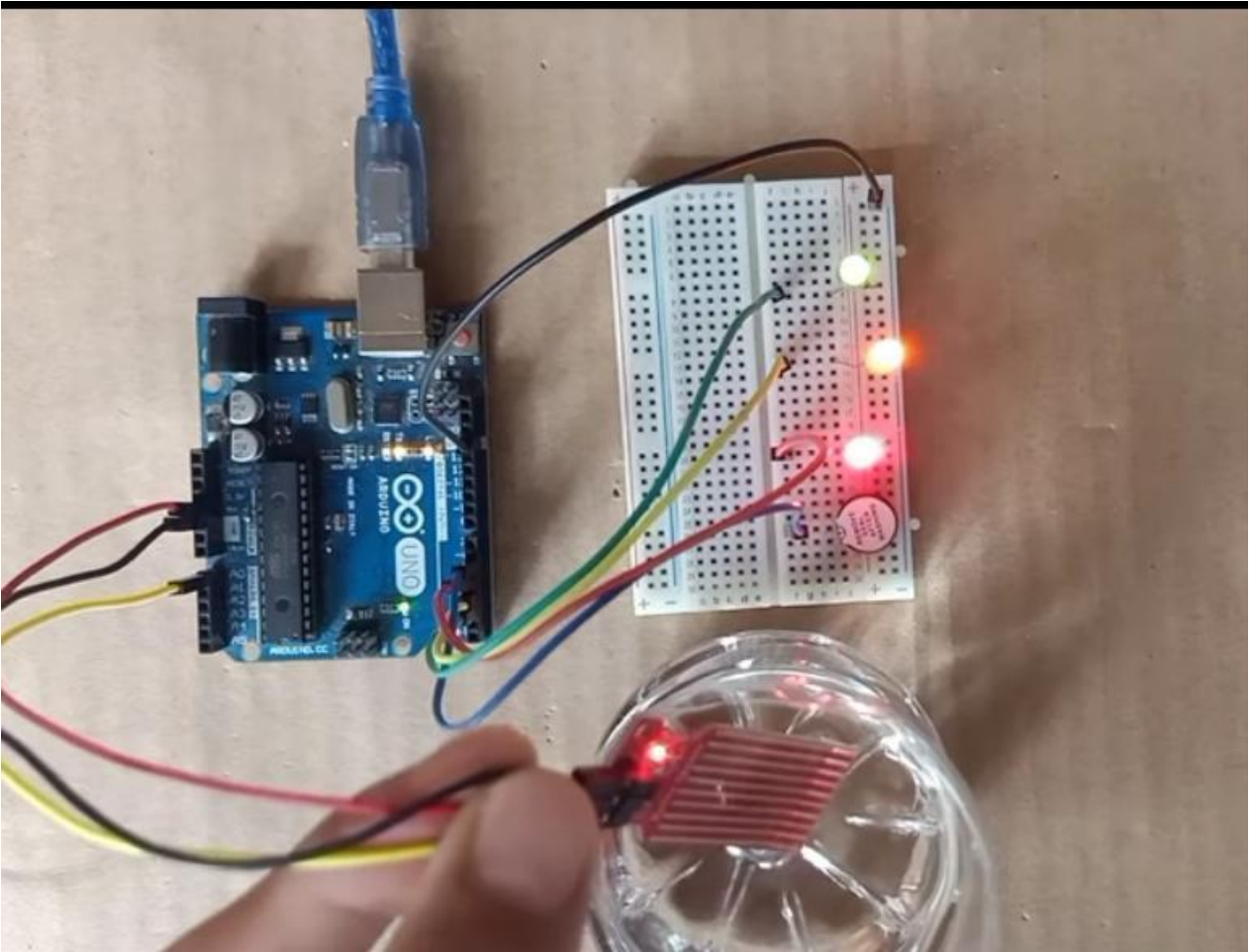
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Introduction

Learn how to make a very simple water level Indicator using Aurdino at home in a few simple steps as follows.....

before starting you should have a basic knowledge of how a transistor work.



Objectives:

1. To learn the working of a water indicator
2. Measure the water level when the circuits indicate when the tank is half and full.
3. To learn how to build simple circuits.

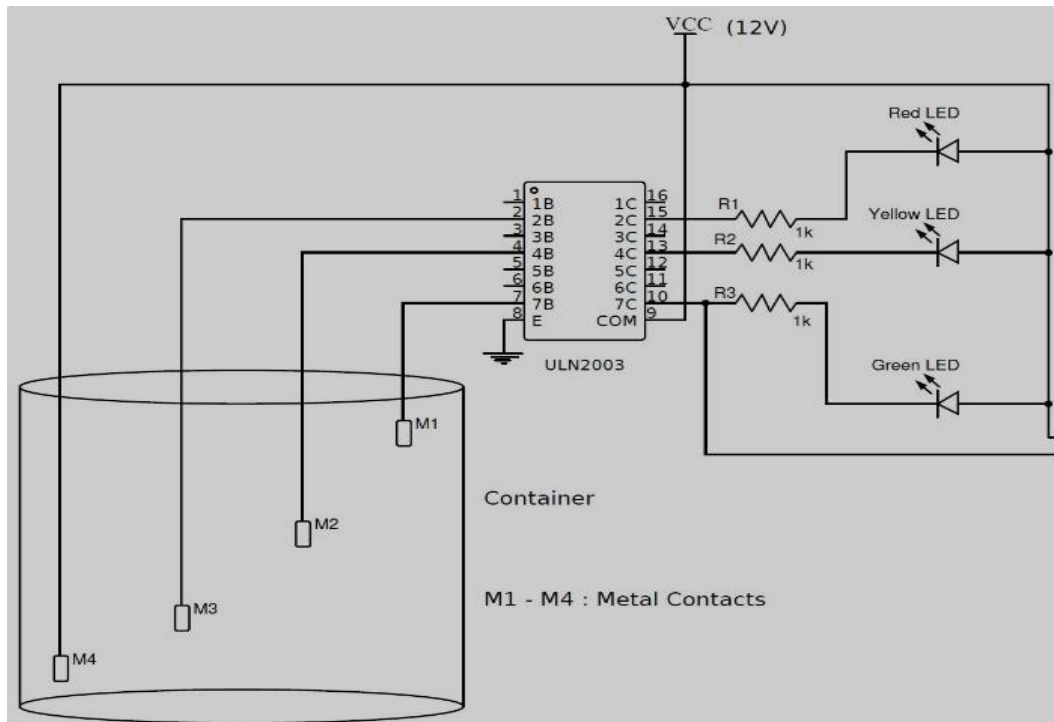
Industry-Based Applications

The water level indicator circuits are used in factories, chemical plants, and electrical substations and in other liquid storage systems. There are many possible uses for this simple system, examples include monitoring a sump pit (to control pump activation), rainfall detection, and leakage detection. Electronic water level circuits have the capability of alerting if there is a water leak somewhere in the factory. When the water level is too high or too low or exceeds the higher limit, it can detect the water level easily by hearing an alarm sound or from different colors of a light bulb. We can also measure the fuel level in motor vehicles and the liquid level containers which are huge in the companies.

Project Methodology

The circuit is designed to indicate three levels of water stored in the tank: low but not empty, half and full but not overflowing. When there is no water in the tank, all the LEDs are off as an indication that the tank is completely empty. When water level increases and touches the sensor, the Red LED will glow indicating that there is water within the tank. As the water level continues to rise and reaches half the tank, Yellow LED will glow. When the water in the tank rises to full an alarm is made by the buzzer as an indication that the tank is full.

Circuit Diagram



Project Procedures.

Components:

1. Arduino Uno



2. Water level sensor module



3. Light-emitting diode (LED's)



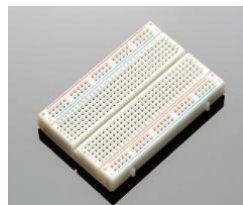
4. 1-Buzzer



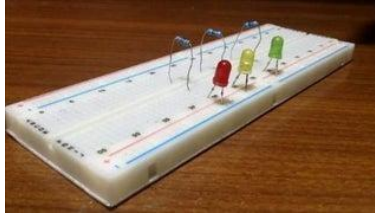
5. Connecting wires



6. Breadboard



Step 1: Assemble LED on Breadboard



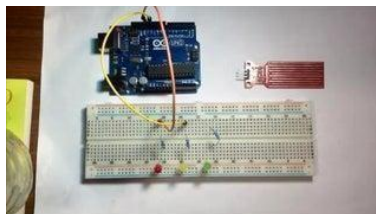
Red: (indicating extremely low level)

Yellow: (indicating half water level)

Green: (indicating full water level)

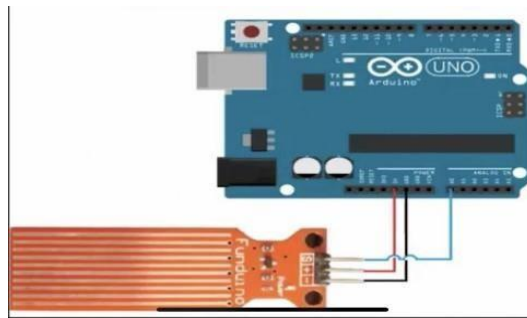
- Connect the cathode of each led to power rail(blue rail) on breadboard which would be the ground supply.
- Connect the anode of LED to different nodes.
- Connect 220 ohms resistor in series with each LED.

Step 2: Make Connections with Arduino and LED Make connections for LED with digital pins on Arduino as follows



- **Red LED** wired to Digital to pin 13
- **Yellow LED** wired to Digital to pin 12 • **Green LED** Wired to Digital to pin 11

Step 4: Connect the Water Sensor with Arduino



Connect the water level sensor to Arduino as follows:

With the Water Sensor pins you need to connect to the Arduino pins

The Negative (-) pin will need connect to the GND on the Arduino with a wire

The Positive (+) pin is needed to connect to the VCC on the Arduino

And The (S) pin needs to be connected to A0 on the Arduino with the wires provided

Advantages

There are many advantages of water level controls, also known also water level indicators, including:

1. POWER SAVER

Living in an age where we need to be more conscious of the energy that we use, a water level controller is ideal at saving power. Normally, regulating water levels can consume electricity and wastewater. However, with automatic controllers, the electricity usage is limited as well as less water needed to regulate supply.

2. MONEY SAVER

A water level controller helps save money by limiting the waste of water and electricity. These devices accurately regulate how much energy is used to protect against any unnecessary water/electricity usage. Over time, the money saved is quite substantial.

3. AUTOMATIC

Another notable advantage with these devices is that they regulate on their own. Eliminating manual operations with a timer switch, the frustrations of manual monitoring water tanks are minimized. Water levels are maintained at the appropriate levels thanks to the automatic operations of these devices.

4. WATER MAXIMIZATION

On average, water pumps are used more during midday. A water level controller can maximize the water usage provided during midday while automatically lessening the water usage at night. This results in an appropriate level of water at all times being maintained, while providing you with the maximum use of your water at the appropriate times.

Conclusion

A Water Level Indicator is a Valuable tool that offers several advantages such as conservation, time and efforts savings, Prevention of damage, and cost – effectiveness. It provides users with real-time information about the water level ,enabling them to make informed decision regarding water usage.However ,it is Important to consider the disadvantages such as the dependence on electronics ,power dependency ,and the need for maintenece .Despite these drawback ,the overall benefits of a water level indicator make it a useful device for managing and monitoring water levels effectively.